

# Open Source for Regulated Manufacturing

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GMP-Ready IIoT Data Logging with Apache IoTDB

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# Slide 1: Open Source for Regulated Manufacturing: GMP-Ready IIoT Data Logging with Apache IoTDB

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Welcome everyone. I'm Lukas Ott, Enterprise Architect, and today I want to show how open source can support scalable, inspectable, and GMP-ready IIoT data platforms.

The key message is simple: compliance does not come from a database alone.

It comes from the validated system around it: controlled ingestion, traceability, access control, retention, monitoring, testing, and lifecycle governance. [.highlight]#This is the practical foundation for GMP-ready IIoT.##



# The Pressure

Connected Manufacturing Meets Regulation





## Slide 2: The Pressure +

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Manufacturing is becoming more connected, more automated, and increasingly data-driven.

Machines, sensors, PLCs, vision systems, and edge devices generate high-frequency data. That data is no longer only useful for dashboards. In regulated manufacturing, machine data can become part of quality decisions, investigations, and inspection evidence.

That creates a tension: we want real-time data and scalable analytics, but we also need trustworthy records, controlled systems, and long-term inspection readiness.

## The Shopfloor Is Becoming a Data Platform



## Slide 3: The Shopfloor Is Becoming a Data Platform

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The shopfloor is no longer a collection of isolated machines. It is becoming a distributed data platform.

But many shopfloor architectures were built machine by machine, site by site, and vendor by vendor. That often leads to fragmented semantics, inconsistent data models, proprietary interfaces, and technology stacks that are difficult to validate end to end.

So the question for today is: how can we build a scalable IIoT data foundation without losing control, traceability, and trust?

## The Core Tension

- Manufacturing needs more real-time data.
- GMP requires trustworthy and inspectable records.
- AI and analytics increase the demand for structured, contextualized data.
- Legacy shopfloor architectures are often fragmented and opaque.

The future shopfloor needs data platforms that are both scalable and inspectable.



## Slide 4: The Core Tension

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This is the core tension of the talk.

On one side, manufacturing wants more connectivity, more automation, more analytics, and eventually AI-supported decision-making.

On the other side, GMP environments require evidence. The system must show where data came from, when it was captured, whether it was changed, who or what changed it, and whether it can be retrieved when needed.



Regulatory Pressure Is Expanding





## Slide 5: Regulatory Pressure Is Expanding

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The pressure is not only more data. It is more connected systems, more cloud, more cybersecurity expectations, and more AI-supported processes.

For regulated manufacturing, the question is no longer only whether a single record is correct. The question is whether the full computerized system lifecycle is controlled.

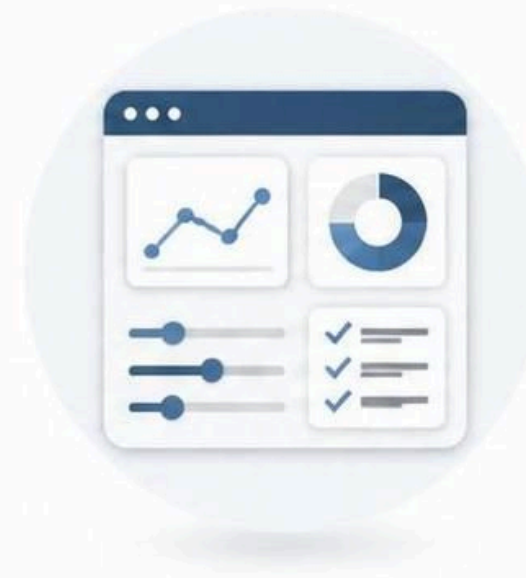
That includes ingestion, storage, processing, access, monitoring, retention, and change management.

## Two Questions Regulators Ask



**Is the  
record  
trustworthy?**

SECURITY • INTEGRITY • TRUST



**Is the  
whole  
system  
controlled?**

GOVERNANCE • OVERSIGHT • CONTROL



## Slide 6: Major Aspects

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For this talk, I simplify the regulatory discussion into two questions.

First: is the electronic record trustworthy?

Second: is the whole computerized system controlled?

Everything else in the architecture maps back to these two questions.

## The Regulatory Lens





# FDA vs. EU in 60 Seconds

**FDA 21 CFR Part 11**



**Electronic records**  
Accurate, reliable, and readily available.



**Secure signatures**  
Unique to one individual and linked to their records.



**Trusted paper-equivalent recordkeeping**  
Electronic records equivalent to paper records.



**Trustworthy records and signatures.**



**EU Annex 11**



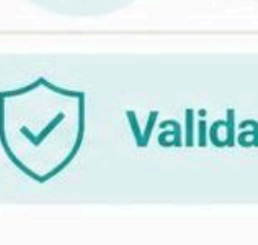
**Validated computerized system**  
Fit for intended use and validated throughout.



**Lifecycle control**  
Requirements managed from design to retirement.



**Governance**  
Clear roles, responsibilities, and oversight.



**Configuration management**  
System changes controlled, tracked, and documented.



**Validated and controlled systems.**

## Slide 8: Part 11 vs. Annex 11 in 60 Seconds

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Those two questions map nicely to two regulatory perspectives.

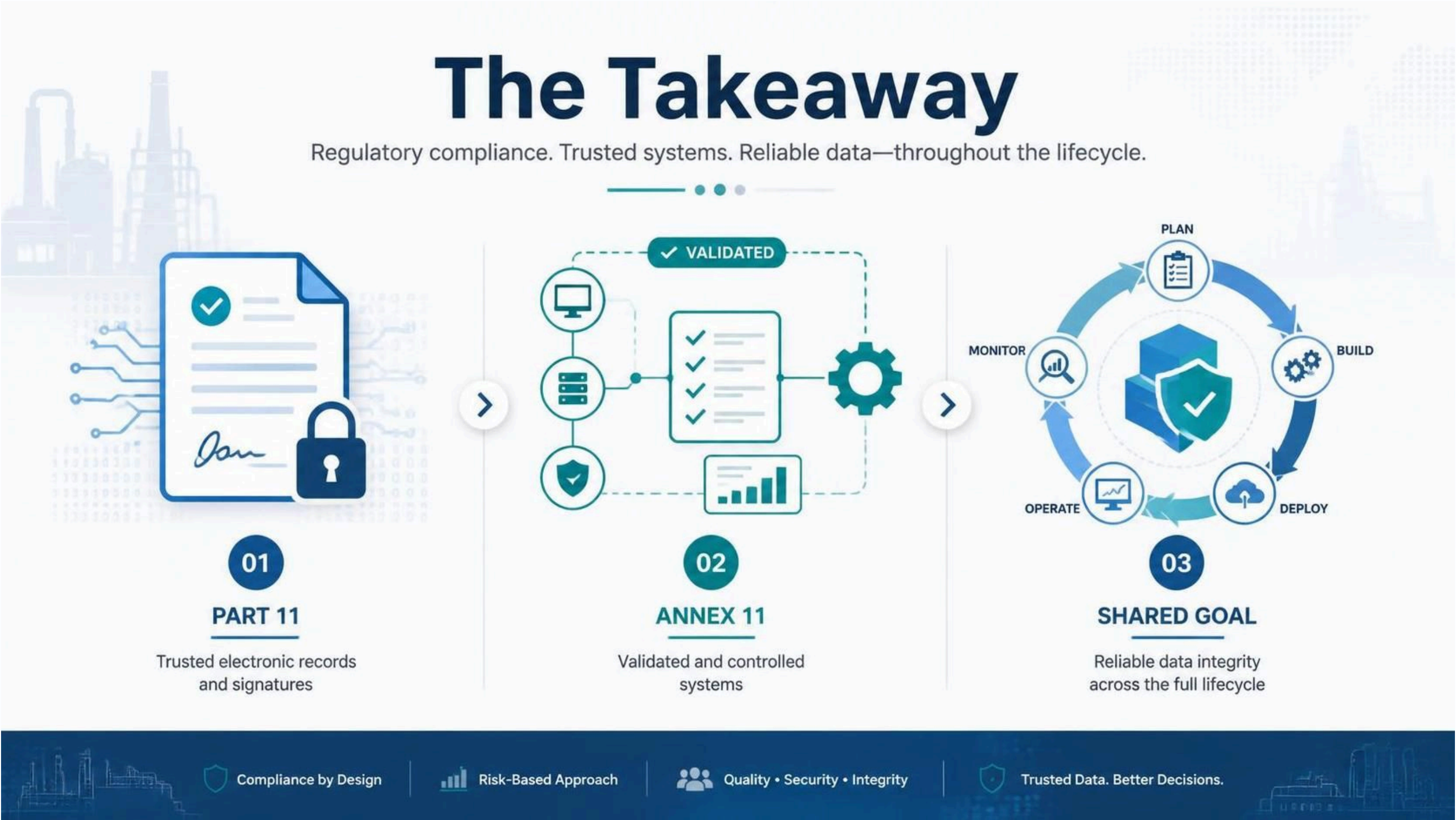
FDA 21 CFR Part 11 focuses on whether electronic records and electronic signatures can be trusted and treated as equivalent to paper-based records.

EU GMP Annex 11 takes a broader lifecycle view. It asks whether the computerized system is validated, governed, secured, monitored, and controlled throughout its lifecycle.

The simplified view is: Part 11 focuses on trusted records and signatures. Annex 11 focuses on validated and controlled systems.



The Shared Compliance Goal



## Slide 9: The Shared Compliance Goal

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The takeaway is simple.

Part 11 focuses attention on trusted electronic records and signatures.

Annex 11 expands the view to the computerized system lifecycle.

Together, they push us toward one shared goal: reliable data integrity from capture to processing, storage, retention, and inspection.

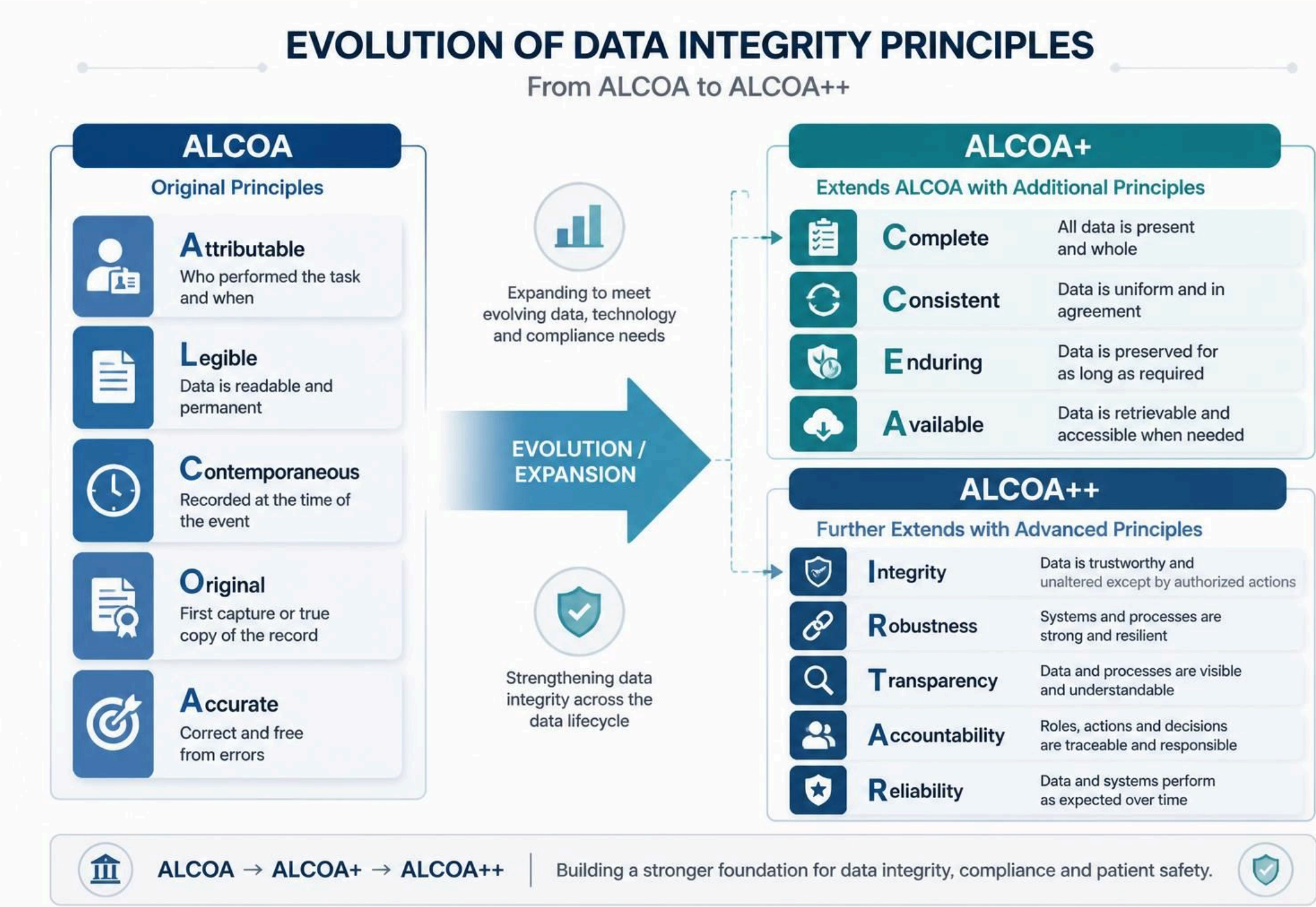


Data Integrity Becomes Architecture





# ALCOA++ as the Practical Lens



## Slide 11: ALCOA++ as the Practical Lens

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ALCOA is often introduced as a data integrity framework: attributable, legible, contemporaneous, original, and accurate.

ALCOA+ extends that with complete, consistent, enduring, and available.

For IIoT, we need to translate those principles into architecture. This is no longer only about a human entering data into a form. It is about machine-generated data, automated ingestion, buffering, synchronization, transformation, storage, and long-term availability.

I do not want to walk through every term on this slide. The important point is the evolution: data integrity becomes an end-to-end platform requirement.



# ALCOA++ Translated to IIoT

For IIoT, ALCOA++ is not a documentation checklist. It is an architecture requirement.

|                                                                                                             |                                                                                                                |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|                                                                                                             |                                                                                                                |
| <b>Attributable:</b> Every reading is linked to a machine, sensor, gateway, service, and timestamp.         | <b>Contemporaneous:</b> Data is captured <b>immediately</b> at the machine event.                              |
| <b>Original:</b> Raw values are preserved <b>before</b> any transformation or aggregation.                  | <b>Complete:</b> Retries, gaps, corrections, and synchronization events remain <b>traceable</b> .              |
| <b>Enduring:</b> Records survive <b>retention periods</b> , infrastructure changes, and recovery scenarios. | <b>Available:</b> Authorized users can retrieve data <b>at any time</b> for operations, review, or inspection. |
| <b>Transparent:</b> Processing logic, configuration, and data movement are auditable.                       | <b>Accountable:</b> System and user actions are tied to <b>responsible identities</b> .                        |

## Slide 12: ALCOA++ Translated to IIoT

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This is where the regulatory language becomes practical.

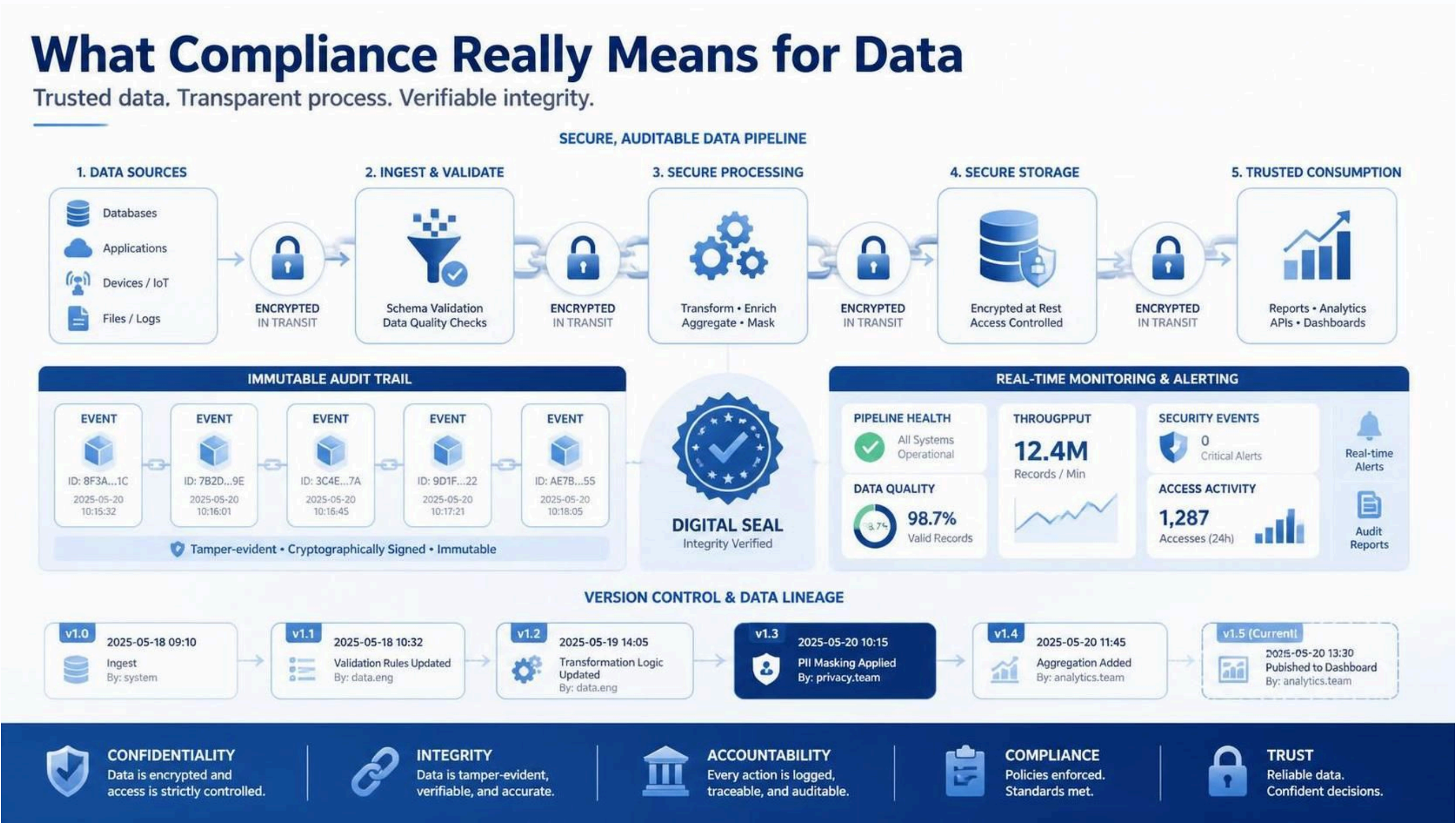
If a sensor reading is attributable, we must know which device, gateway, and service produced it.

If the data is contemporaneous, timestamp strategy matters.

If the data is original, we need to preserve raw values before downstream transformation.

If the data is complete, retry behavior, synchronization gaps, and corrections must be visible.

This is why data integrity has to be designed into the platform.





## Slide 13: Compliancy and Data

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This slide translates compliance into a data pipeline.

It is not only encrypted storage. It is a controlled data journey: validated ingestion, protected processing, secure storage, traceable consumption, audit trails, lineage, monitoring, and alerting.

At every stage we need to answer: where did the data come from, what happened to it, who or what accessed it, and can we prove that later?

# What Can Go Wrong on the Shopfloor?

|                                         |                                          |
|-----------------------------------------|------------------------------------------|
|                                         |                                          |
| Network outage during production        | Duplicate readings after retry           |
| Missing or inconsistent timestamps      | Unauthorized data access                 |
| Silent data loss during synchronization | Uncontrolled transformation logic        |
| Retention or backup failure             | Configuration drift between environments |

A GMP-ready IIoT architecture must be designed for failure, not only for normal operation.

## Slide 14: What Can Go Wrong on the Shopfloor?

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This is important because real manufacturing environments are not perfect.

Networks fail. Gateways restart. Messages are retried. Timestamps can differ. Configuration can drift. Data can be transformed without enough traceability.

A credible IIoT data architecture must handle these failure modes in a controlled and observable way.



## Why Open Source?





# Why Open Source Helps Validation



## Slide 16: Why Open Source Helps Validation

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Open source does not remove validation work. It does not make a system compliant by default.

But open source gives us something valuable in regulated environments: transparency, inspectable architecture, open interfaces, community review, and less dependency on a single vendor roadmap.

For regulated industries, that matters because validation depends on understanding and controlling the implemented system.



## Open Source Is Not Automatically Compliant

- The software is not validated by default.
- The implemented system must be qualified and validated.
- Supplier and community governance still need assessment.
- Security patching and dependency management must be controlled.
- Documentation, testing, SOPs, and change control remain mandatory.

Open source is not a shortcut around GMP. It is a transparent foundation that can support validated architectures.

## Slide 17: Open Source Is Not Automatically Compliant

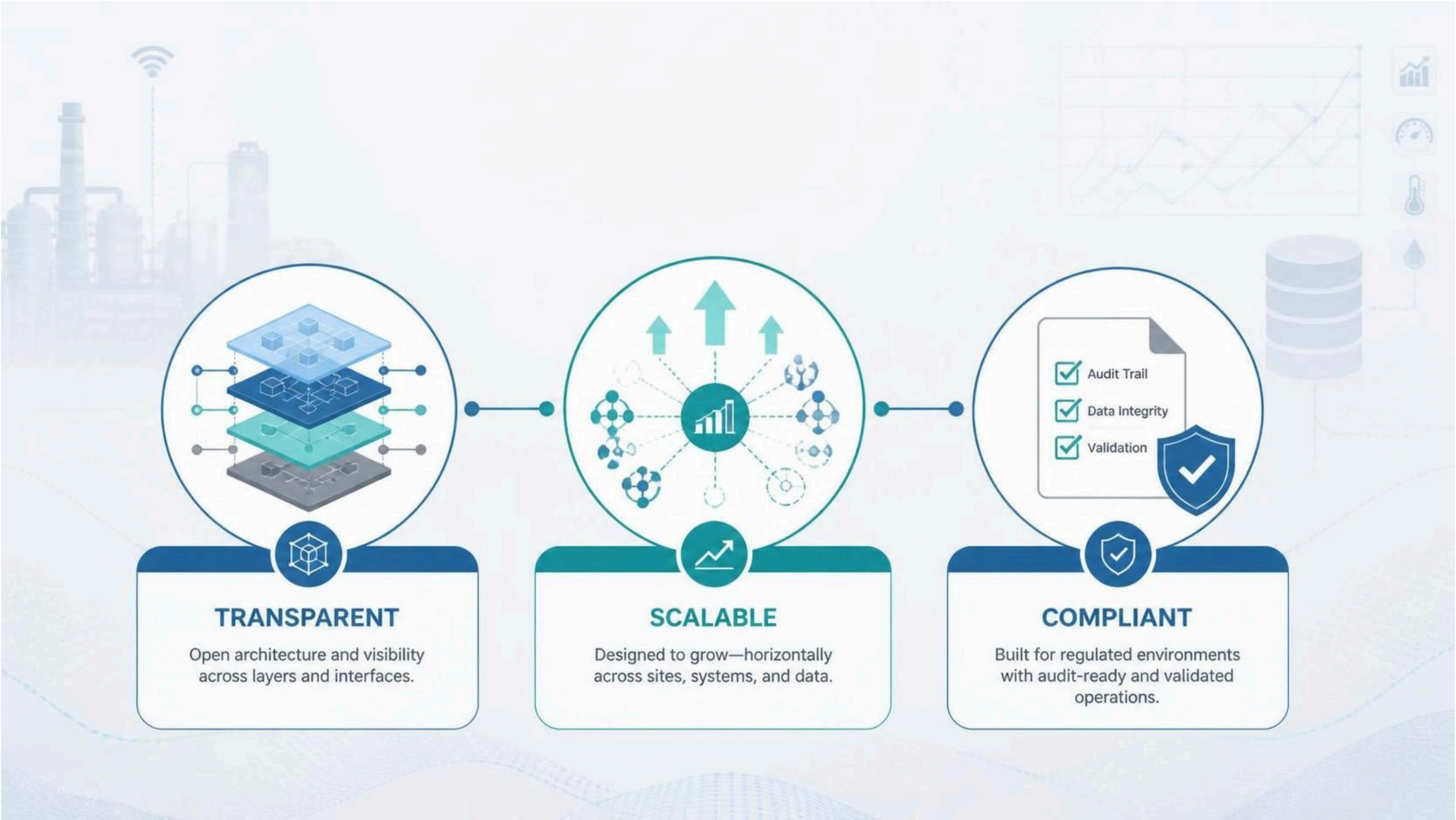
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This is an important distinction.

We should not say that an open source component is automatically compliant. Compliance comes from the implemented and validated system.

But transparency, open standards, and inspectable behavior can make technical assessment, integration, and long-term governance easier.

# What to Remember About Open Source





## Slide 18: What to Remember About Open Source

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If there are three things to remember about open source in this context, they are transparency, scalability, and validatability.

Transparency means open interfaces and inspectable architecture.

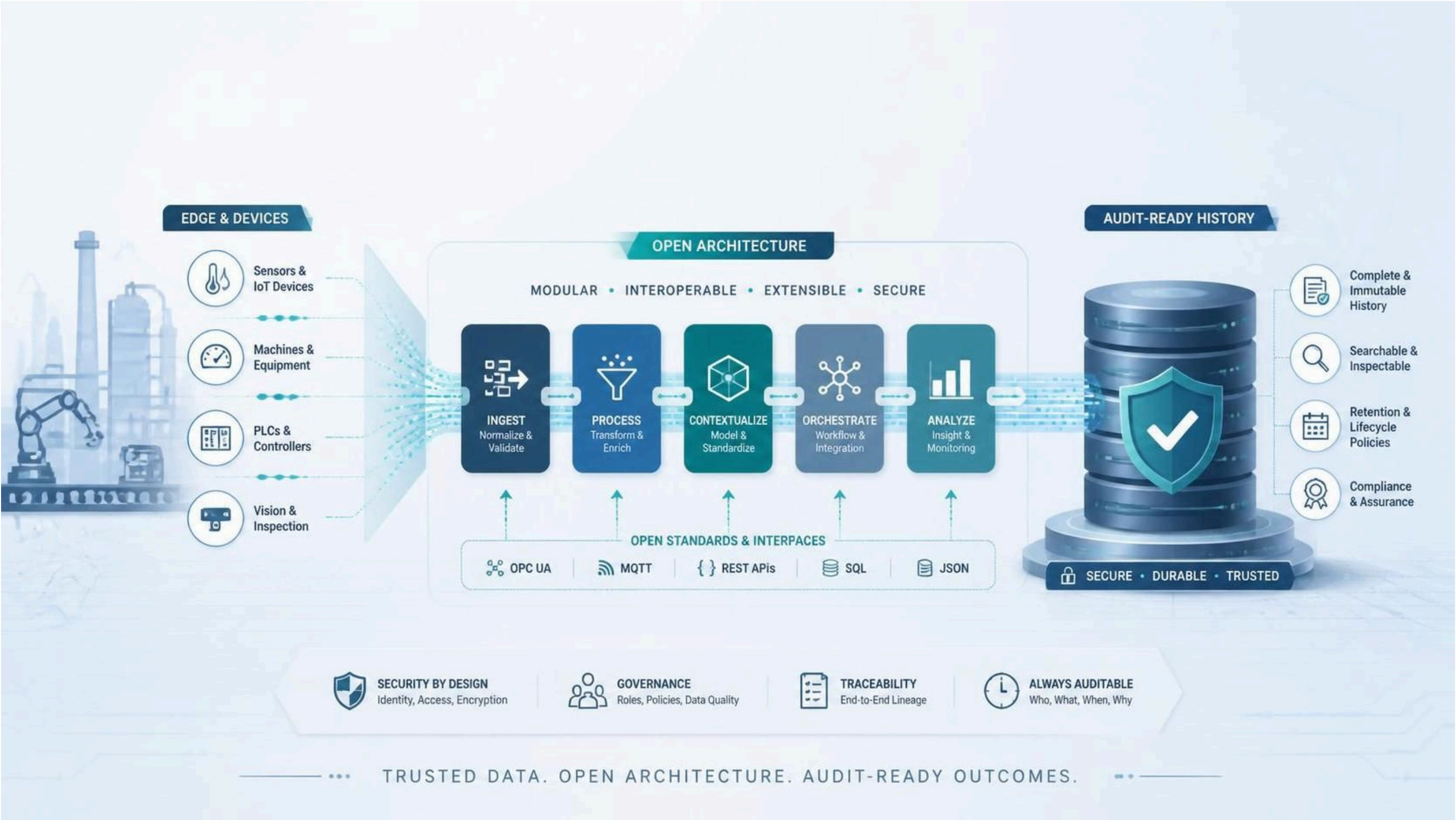
Scalability means the architecture can grow across machines, sites, and data volumes.

Validatability means the system can be specified, tested, monitored, documented, and controlled.

From Open Architecture to Apache IoTDB



# Open Architecture for Audit-Ready History





## Slide 20: Open Architecture for Audit-Ready History

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Before we look at individual technologies, this is the architecture pattern.

We start with edge devices, machines, PLCs, sensors, and inspection systems.

We ingest and validate signals. We process and contextualize data. We orchestrate workflows. We analyze and visualize information.

The outcome must be audit-ready history: searchable, inspectable, retained, and governed.

The bottom layer is critical: security, governance, traceability, and auditability are not add-ons. They must run across the full architecture.

## Example Intended Use

- Capture high-frequency machine and sensor readings.
- Buffer data locally when central connectivity is unavailable.
- Synchronize data into a historian without duplicates.
- Visualize current and historical data.
- Support traceability, operational resilience, and inspection readiness.

The goal is not to validate a database in isolation. The goal is to validate the implemented system for its intended use.

## Slide 21: Example Intended Use

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In regulated environments, intended use is central.

For this example, the intended use is to capture, buffer, store, synchronize, and visualize high-frequency machine data from regulated manufacturing equipment.

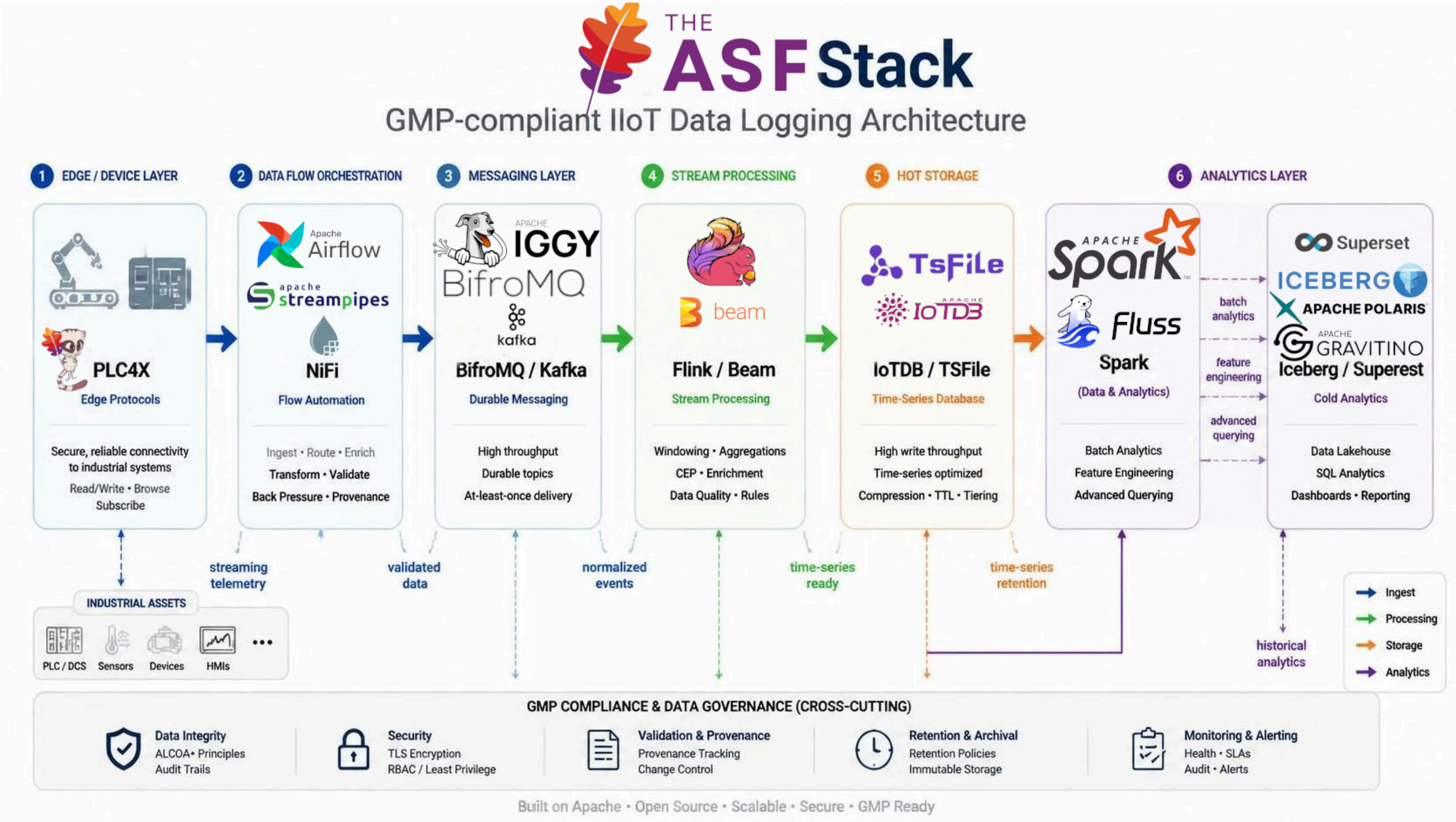
The system supports operational monitoring, data integrity, and inspection readiness.

It does not replace the full quality management system, electronic batch record approval, or final product release process.



# Apache-Based IIoT Data Logging

Reference Architecture



## Slide 22: Apache-Based IIoT Data Logging +

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This slide is not saying every project is mandatory. It shows a modular Apache-based reference architecture.

For this talk, the core is Apache IoTDB and TSFile as the historian foundation.

Apache PLC4X can help connect to industrial protocols.

Apache NiFi can orchestrate controlled data flows.

Kafka or BifroMQ can provide buffering and decoupling.

Flink or Beam can process streams.

Iceberg, Spark, and Superset can support analytical consumption and visualization.

The most important part is the cross-cutting layer: data integrity, security, validation evidence, retention, monitoring, and alerting.

## Apache IoTDB as the Historian Core

- Optimized for high-frequency industrial time-series data.
- Stores structured device and sensor measurements.
- Supports integration with authentication, authorization, encryption, monitoring, and retention controls.
- Can serve as the historian core inside a validated IIoT architecture.
- Works together with controlled ingestion, metadata, logging, and validation processes.



## Slide 23: Apache IoTDB as the Historian Core

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Apache IoTDB is the time-series database at the center of the architecture.

The important wording here is that IoTDB can serve as the historian core inside a validated architecture.

The database alone does not make the system compliant. The system becomes GMP-ready when the architecture around it provides controlled ingestion, traceability, access control, retention, monitoring, testing, and lifecycle governance.

## Apache TSFile as the Storage Foundation

- Time-series file format optimized for efficient storage and query.
- Supports append-oriented storage patterns.
- Helps separate raw time-series history from downstream analytics.
- Supports reproducibility and inspection when combined with controlled storage, access, and retention design.
- Can be paired with analytical formats such as Apache Iceberg.

## Slide 24: Apache TSFile as the Storage Foundation

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TSFile is the underlying time-series file format used by IoTDB.

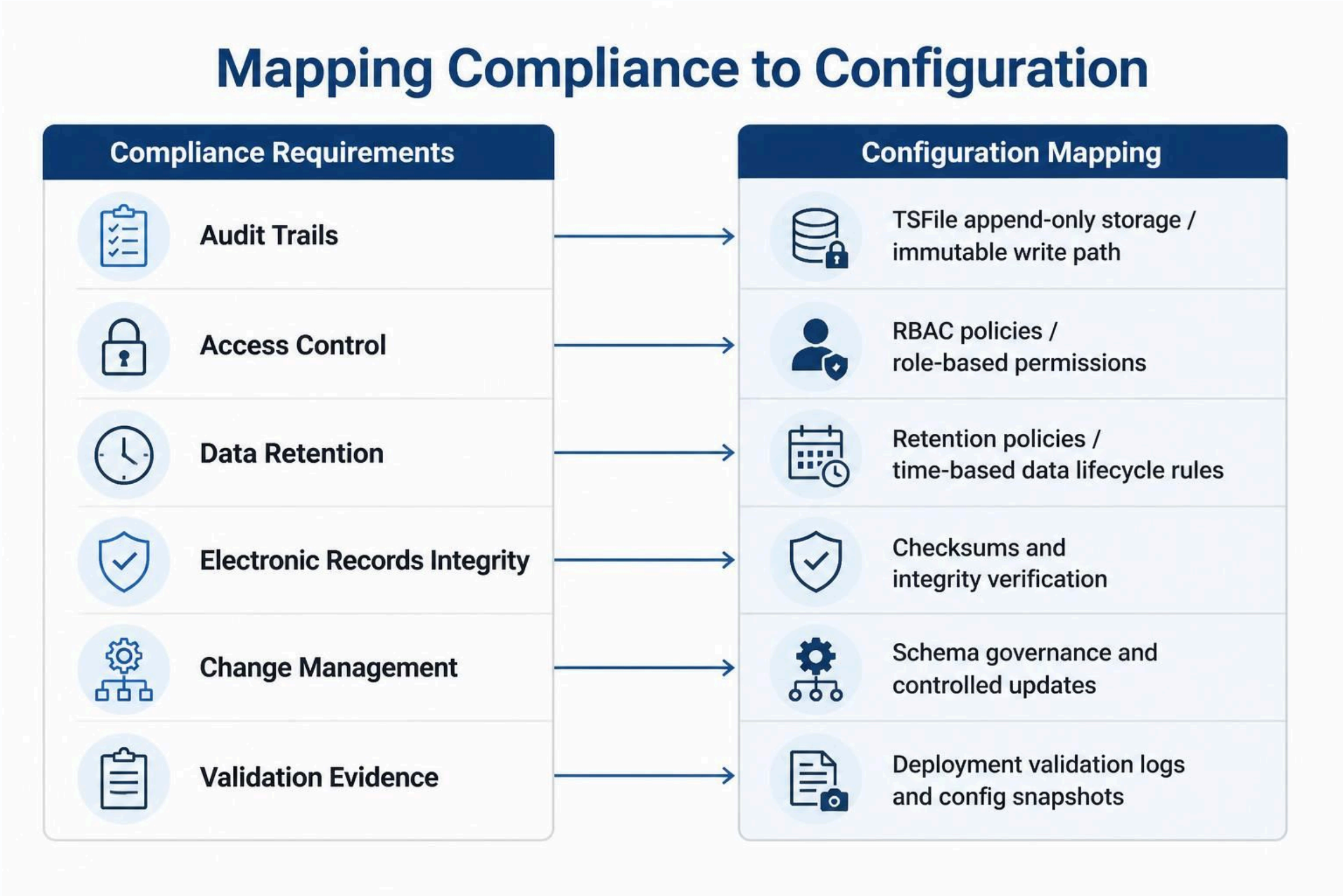
Append-oriented storage patterns are useful because they support preservation of historical data and reproducibility.

However, immutability is not only a file-format property. A tamper-evident and inspection-ready design also depends on access control, storage policies, backups, checksums, monitoring, and operational procedures.



# Mapping Compliance

Expectations to Technical Controls



## Slide 25: Mapping Compliance +

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This is where compliance becomes engineering.

A requirement like record integrity is implemented through controlled write paths, checksums, raw data preservation, and access control.

A requirement like validation evidence becomes deployment logs, test records, configuration snapshots, monitoring records, and change history.

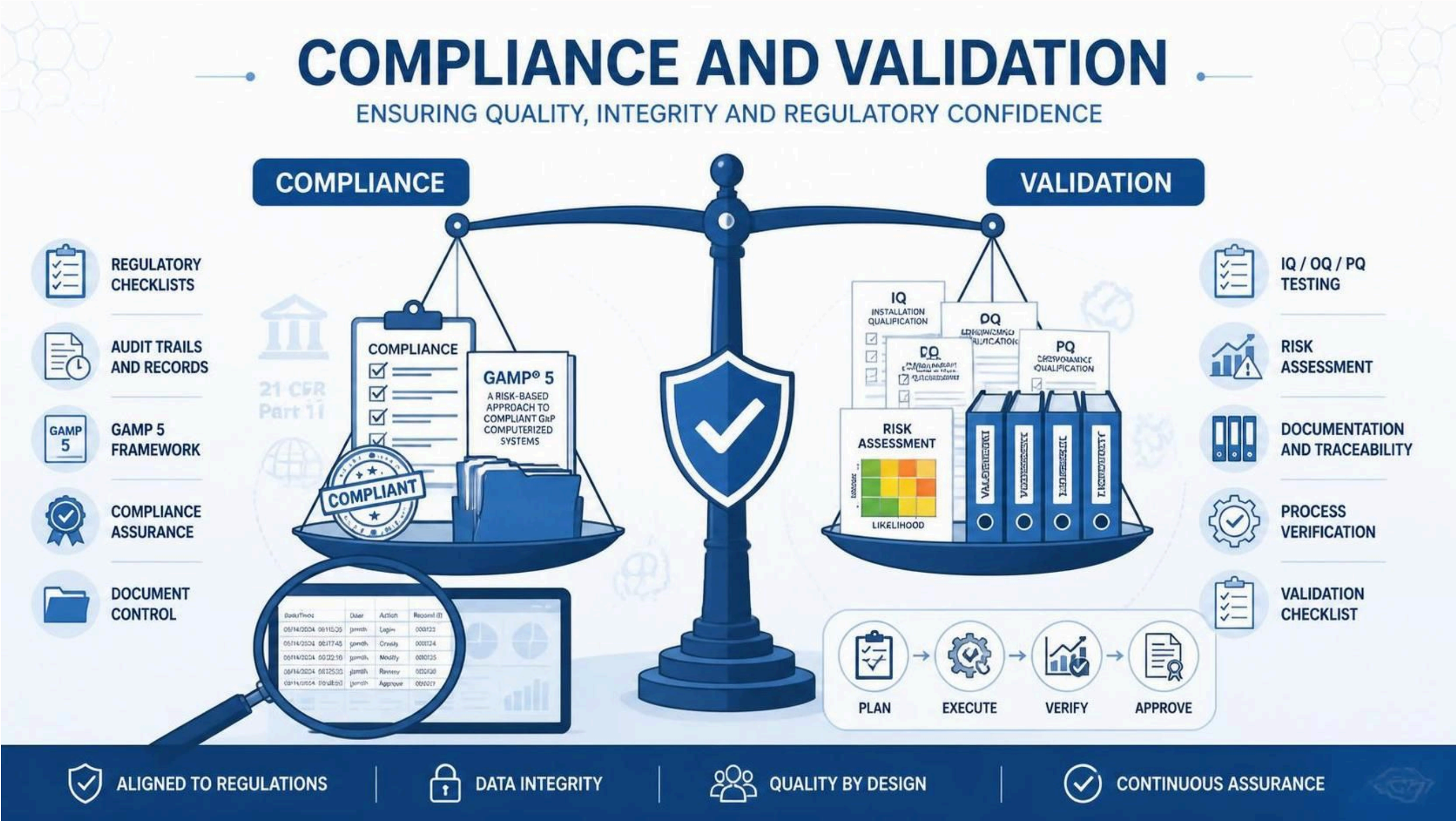
The key is that every compliance expectation should map to a concrete architecture or operational control.



From Architecture to Validation Evidence









## Slide 27: Compliance and Validation

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After the architecture, the next question is: how do we make this usable in a regulated environment?

The answer is not one feature. It is a validation lifecycle.

That lifecycle includes intended use, system boundaries, risk assessment, user requirements, design specifications, testing, documented evidence, operating procedures, monitoring, periodic review, and controlled change.

## Validation Starts with System Boundaries

- What is GMP-relevant?
- What is the regulated record?
- Which components create, modify, store, or display the record?
- Which integrations are in scope?
- Which users, services, and systems can access the data?
- What is the intended use?
- What is explicitly out of scope?

## Slide 28: Validation Starts with System Boundaries

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Before validating anything, we need to define the system boundary.

For IIoT this is especially important because the architecture can include machines, gateways, brokers, stream processors, databases, dashboards, cloud services, and analytical platforms.

Not every component has the same GMP relevance, but every component that creates, modifies, stores, or displays regulated records needs to be understood.



## Risk-Based Validation with GAMP 5

1. **Intended Use**: Define what the system is expected to do.
2. **Risk Assessment**: Identify patient, product quality, and data integrity risks.
3. **User Requirements**: Define business, quality, and operational requirements.
4. **Design Specifications**: Translate requirements into controlled architecture and configuration.
5. **IQ / OQ / PQ**: Prove installation, operation, and performance.
6. **SOPs and Training**: Ensure controlled and repeatable use.
7. **Monitoring and Review**: Verify ongoing compliant operation.
8. **Change Control**: Manage updates without breaking validated state.

## Slide 29: Risk-Based Validation with GAMP 5

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The key point is that we do not validate open source differently from closed source software.

We validate the implemented system based on intended use, risk, and evidence.

GAMP 5 gives us a practical risk-based approach for doing that.

# What Good Testing Looks Like

| Test | What it proves                                             | IIoT example                                                                       |
|------|------------------------------------------------------------|------------------------------------------------------------------------------------|
| IQ   | The system is installed repeatably and correctly.          | Correct versions, configuration, certificates, storage paths, deployment scripts.  |
| OQ   | The system functions as specified.                         | RBAC, retention, backup, synchronization, failover, audit logging, monitoring.     |
| PQ   | The system performs under realistic production conditions. | Peak sensor load, network outage, replay, idempotent sync, long-running ingestion. |



## Slide 30: What Good Testing Looks Like

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IQ means Installation Qualification. It proves that the system is installed correctly and repeatably.

OQ means Operational Qualification. It proves that specified functions behave as intended.

PQ means Performance Qualification. It proves that the system performs under realistic operating conditions.

For an IIoT historian, PQ is especially important because peak load, network interruption, retry behavior, and synchronization can directly affect data completeness and trust.

# Main Elements of 21 CFR Part 11

21 CFR Part 11 requirements for trustworthy electronic records and signatures.



## VALIDATION

Ensure systems are fit for intended use and consistently perform as expected.



## SECURITY

Protect records and signatures to ensure confidentiality, integrity, and availability.



## RECORD RETENTION

Retain records for the required period and ensure accurate, readable reproduction.



## AUDIT TRAILS

Create secure, computer-generated audit trails that record time-stamped actions and changes.



## ELECTRONIC SIGNATURES

Use electronic signatures that are unique to each individual and linked to their records.



## TRAINING

Ensure personnel are trained on Part 11 requirements and their role in compliance.

## Slide 31: Part 11 Control Themes

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Part 11 brings recurring control themes: validation, security, record retention, audit trails, electronic signatures, and training.

Not every historian use case requires electronic signatures inside the historian itself. In many architectures, signatures are handled in a higher-level MES, QMS, LIMS, or review workflow.

But records, access control, retention, auditability, training, and validation evidence must still be addressed in the implemented process.



## The Demo: Resilience and Traceability





## Demo Scenario

### Network Outage Without Data Loss

- A production line continues generating readings.
- The central historian is temporarily unavailable.
- The local system continues accepting data.
- Readings are buffered safely.
- Synchronization resumes when connectivity returns.
- Duplicate records are avoided through idempotent behavior.
- Historical data becomes visible in Apache IoTDB.

The demo proves continuity, traceability, and recoverability under realistic shopfloor conditions.

## Slide 33: Demo Scenario +

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The demo focuses on one realistic shopfloor problem: connectivity is not perfect.

A production line may continue generating readings even when the central historian is temporarily unavailable.

The system must keep accepting data locally, buffer it safely, and synchronize later without duplicates or silent data loss.

That is the pattern I want to show now.

## Live Demo: Hybrid IoT Data Logging and Visualization

- Goal: show the path from ingestion to buffering to synchronization.

## Slide 34: Live Demo: Hybrid IoT Data Logging and Visualization

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During the demo, focus on the control points.

First, authenticated ingestion.

Second, local buffering.

Third, synchronization progress.

Fourth, historical visibility in IoTDB.

Fifth, idempotent resume behavior, meaning that repeating the operation does not create duplicates or incorrect state.



# Demo Steps

| Step | Action                                                                     |
|------|----------------------------------------------------------------------------|
| 1    | Log in as <code>operator</code> with <code>operator</code> .               |
| 2    | Send sample readings to <code>/ingest</code> or use the ingestion helper.  |
| 3    | Open the dashboard and observe the local buffer chart.                     |
| 4    | Trigger <code>Sync to IoTDB</code> and watch the live SSE progress stream. |
| 5    | Verify that synchronized historical data appears in the IoTDB chart.       |

## Slide 35: Demo Steps

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If time is short, run only ingestion and synchronization, then point to the dual-chart comparison and recovery behavior.

SSE means Server-Sent Events.

Idempotent means repeating the action does not create duplicates or incorrect state.

## Conclusion





From Edge Data to Audit-Ready History





## Slide 37: From Edge Data to Audit-Ready History

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The future of regulated manufacturing will be more connected, more automated, and more data-driven.

But the core questions remain simple.

Can we trust the record?

Can we prove the system is controlled?

Apache IoTDB and the Apache ecosystem give us a strong open foundation for high-frequency IIoT data logging.

But compliance is created by the full validated system: architecture, controls, evidence, monitoring, procedures, and change control.

Open source is not a shortcut around GMP. It is a transparent foundation we can validate and evolve together.

## Final Takeaways

- Regulated manufacturing does not just need more data.
- It needs data that can be trusted, inspected, and preserved.
- Apache IoTDB can serve as the historian core of a GMP-ready IIoT architecture.
- Apache technologies provide modular building blocks from edge to analytics.
- Validation depends on intended use, risk, evidence, procedures, and lifecycle control.
- Open source supports transparency, scalability, and validatability.

Open source is not a shortcut around validation. It is a foundation for transparent, scalable, and inspectable manufacturing data platforms.

## Slide 38: Final Takeaways

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If there is one message to remember, it is this:

Apache IoTDB can serve as the historian core of a GMP-ready IIoT architecture, but compliance does not come from the database alone.

It comes from the validated system around it: controlled ingestion, traceability, access control, retention, monitoring, testing, and lifecycle governance.

## Q&A

- Thank you.
- Projects:
  - Apache IoTDB: <https://iotdb.apache.org/>
  - Apache PLC4X: <https://plc4x.apache.org/>
  - Apache StreamPipes: <https://streampipes.apache.org/>
- Contact:
  - Lukas Ott <[otluk@apache.org](mailto:otluk@apache.org)>



## Slide 39: Q&A;

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Thank you for your attention. I'm happy to take questions.